



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
15-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

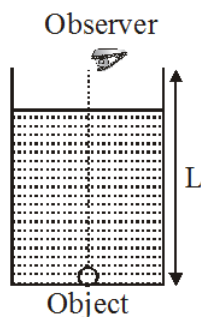
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A cylinder is filled with a liquid of refractive index μ . The radius of the cylinder is decreasing at a constant rate K . The volume of the liquid inside the container remains constant at V . The observer and the object O are in a state of rest and at a distance L from each other. The apparent velocity of the object as seen by the observer, is : (when radius of cylinder is r)



(A) $\frac{(1 - \mu)2KV}{(\pi\mu r^3)}$

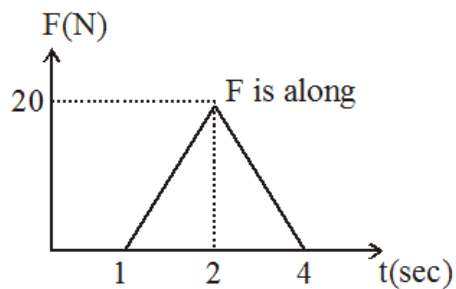
(B) $\frac{(1 - \mu)2KV}{(\pi\mu Lr^2)}$

(C) $\frac{(1 - \mu)2K}{\mu}$

(D) $\frac{(1 - \mu)K}{2\mu}$

Space for Rough Work

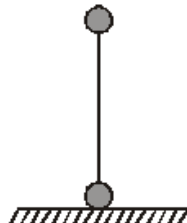
2. A particle of mass 2 kg is moving along x-axis with a velocity of 5 m/s. It crosses the origin at $t = 0$. A time varying force whose variation is as shown starts acting on a particle at $t = 1$ sec. For this situation, mark out the correct statement.



- (A) The particle will come to rest instantaneously at $t = 2$ sec
 (B) The final velocity of the particle will be 18 m/s.
 (C) Work done by the force F on the particle is 375 J.
 (D) All of the above

Space for Rough Work

3. Two tennis balls of mass 60 g are attached with a massless rubber thread, and held in the vertical position as shown in the figure. In this position, the unstretched length of the rubber thread is 40 cm. The upper ball is slowly raised vertically upward, until the lower ball just becomes unsupported by the ground. At this time the length of the thread is 1m. The rubber thread exerts a force which is proportional to its extension. Choose correct statement.



- (A) Work done by external agent while the upper ball was raised is 0.6J.
 (B) Work done by external agent while the upper ball was raised is 0.23J.
 (C) Releasing the upper ball, the speed which it hits the lower one is 5.1 m/s.
 (D) The time that elapses between the release of the upper ball and the collision is 0.5 sec.
4. A mass of $m = 1\text{kg}$ moves initially with a velocity of $v_0 = 10\sqrt{15}\text{ m/s}$. A constant power $P = 1000\text{ Watt}$ is applied so as to increase its velocity. Find the time that elapses before the acceleration is reduced to half of its original value.
- (A) 4 sec (B) 9/4 sec (C) 1 sec (D) 4.5 sec

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

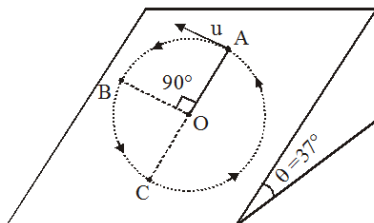
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

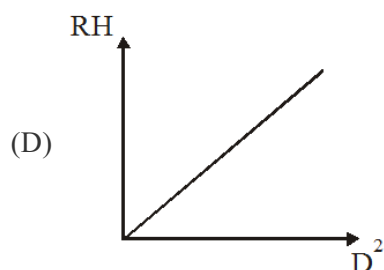
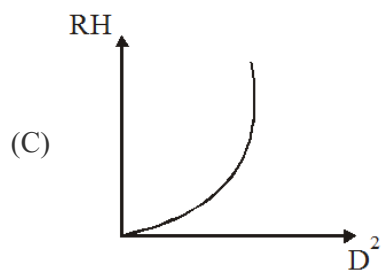
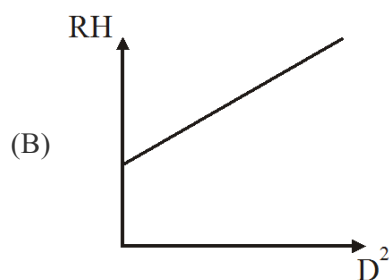
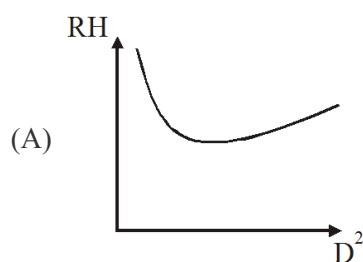
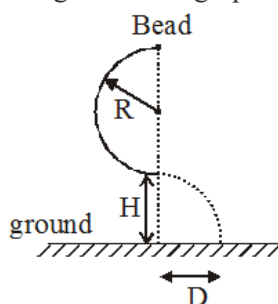
5. A small sphere of mass m is connected by a string to a nail at O and moves in a circle of radius r on the smooth plane inclined at an angle θ with the horizontal. If the sphere has a velocity u at the top position A . Mark the correct option(s) :-



- (A) Minimum velocity at A so that string does not get slack instantaneously is $\sqrt{\frac{3}{5}gr}$.
- (B) Tension at B is $9mg/5$ if the sphere has required velocity as in option(A).
- (C) Tension at C is $18mg/5$ in situation of option(B).
- (D) None of these

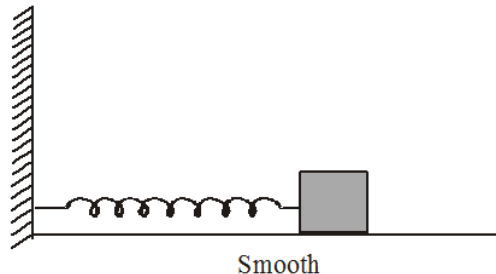
Space for Rough Work

6. A semicircular wire of radius R is oriented vertically. A small bead is released from rest from the top of the wire. It slides without friction under the influence of gravity to the bottom, where it then leaves the wire horizontally and falls distance H to the ground. The bead lands a horizontal distance D away from where it was launched. Which of the following is correct graph of RH versus D^2 ?



Space for Rough Work

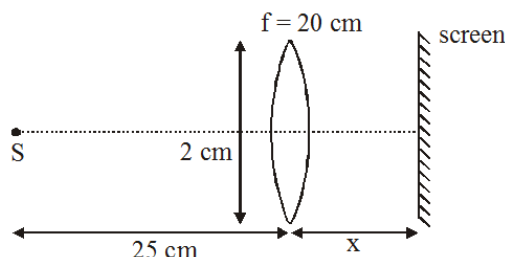
7. A block of mass 1 gm carrying a charge $1\mu\text{C}$ is connected to a spring of stiffness $(\pi^2 \times 10^{-3}) \text{ N/m}$ whose other end is fixed to a wall. The block rests on a smooth ground as shown. By making a suitable arrangement, an electric field lying in the plane of the figure, directed towards right with strength $\left(\frac{\pi}{10}\right) \text{ MV/m}$ is switched on exactly for an interval of 1 ms after every half second. Take the moment when the field is switched on for first time as $t = 0$. Now, select the correct options.



- (A) The maximum extension in the spring is 0.1414 m (nearly)
- (B) The spring never gets compressed.
- (C) The time taken by the block to start from mean position and reach the position of maximum elongation of spring is 0.75s.
- (D) The time taken by the block to start from mean position to the position of maximum elongation of spring is 0.5 s.

Space for Rough Work

8. A point source of light S is placed on the axis of the lens of focal length 20 cm as shown in the figure. A screen is placed normal to the axis of the lens at a distance x from it. Treat all rays as paraxial. Then which of the following is/are correct regarding the intensity of the image produced on the screen of the point source ?



- (A) As x is increased from zero, intensity continuously decreases.
 (B) As x is increased from zero, intensity first increases then decreases.
 (C) Intensity at centre of screen for $x = 90$ cm and $x = 110$ cm is same.
 (D) Radius of bright circle obtained on screen is equal to 2 cm for $x = 300$ cm.
9. A concave mirror forms real image of a point object 'O' lying on the optical axis at a distance of 60 cm from the pole of mirror. The focal length of mirror is 30 cm. Now, the mirror is cut into two halves symmetrically. In figure-1 both halves are shifted at a distance of 2 mm apart in a direction perpendicular to the optical axis. In figure-2 both halves are rotated (before being separated) about 'O' with reference to original position. Let I_1 and I_2 are the image formed by both halves in both figure. Choose the correct statement(s).

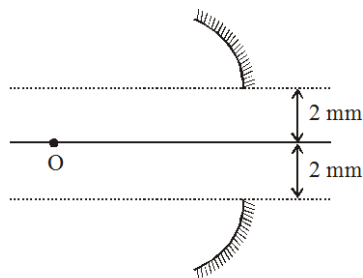


Figure-1

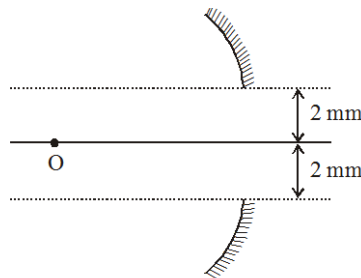
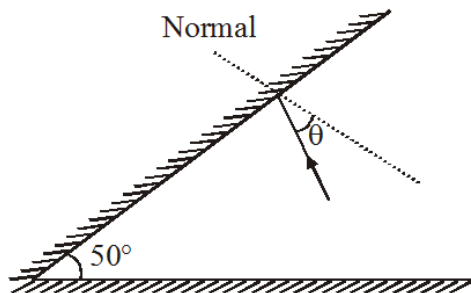


Figure-2

- (A) In figure-2, I_1 and I_2 are coincide.
 (B) In figure-1, I_1 and I_2 are separated.
 (C) In figure-1, the distance between I_1 and I_2 is 8 mm.
 (D) In figure-2, the distance between I_1 and I_2 is 4 mm.

Space for Rough Work

10. Two plane mirrors are placed in such a way that they make an angle 50° as shown in figure. A ray of light falls on one mirror and it reflects from each mirror once. For the given situation mark the correct statement(s).



- (A) The value of θ must be less than 40° .
 (B) θ can have any value ranging from 0° to 90° .
 (C) Total deviation produced in light ray is independent of value θ .
 (D) Laws of reflection are valid for only plane and spherical mirrors.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

An experiment is performed to calculate the focal length of a convex mirror using convex lens. The setup consists of two pins, optical bench, one convex lens and one convex mirror. When only convex lens is placed at a distance 60 cm from pin (object), its image is formed at a distance 30 cm from the lens as shown in figure-(i). Now convex mirror is also kept right to the lens without changing distance between pin (object) and lens. The mirror is kept at a distance 10 cm from lens such that position of image of object pin and position of object pin coincides as shown in figure-(ii)

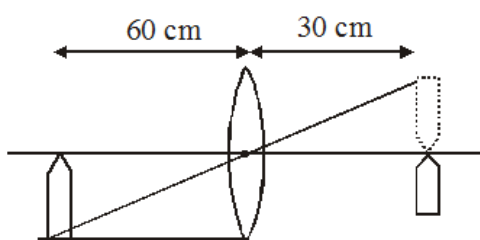


Figure-(i)

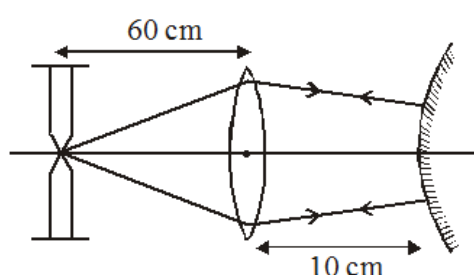


Figure-(ii)

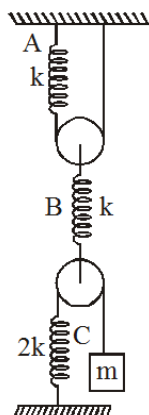
Now answer the following questions.

1. Focal length of convex lens used (in cm) is :
2. Focal length of convex mirror (in cm) is :

Space for Rough Work

Paragraph for Questions 3 and 4

In the given figure, the block is attached with a system of three ideal springs A, B, C. The block is displaced slightly by a distance 'x' from its equilibrium position vertically downwards and released.

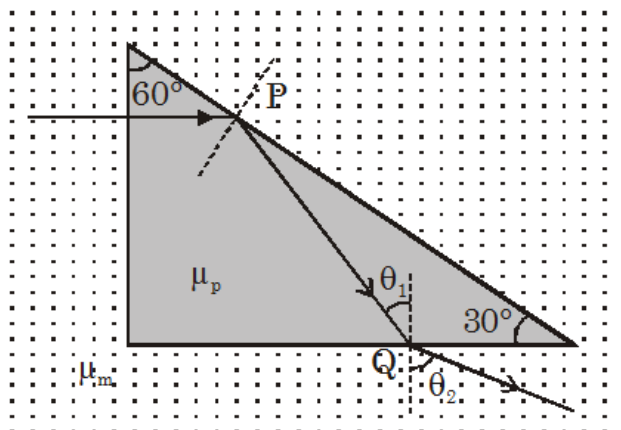


3. The ratio of deformation of spring B to spring C is
4. The time period of small vertical oscillations of block is $2\pi\sqrt{a\frac{m}{k}}$. Find a

Space for Rough Work

Paragraph for Questions 5 and 6

A ray of light is incident normally on one face $30^\circ - 60^\circ - 90^\circ$ prism of refractive index $\mu_p = 5/3$ immersed in water of refractive index $\mu_m = 4/3$ as shown in the figure.



5. The exit angle θ_2 of the ray is $\sin^{-1} \left(\frac{x}{16} \right)$, find the value of x .
6. The minimum refractive index of surrounding medium so that total internal reflection at point P ceases is $\frac{n}{\sqrt{3}}$, find the value of n .

Space for Rough Work

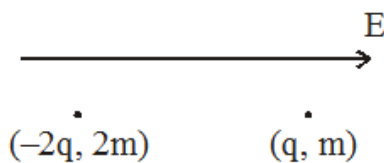
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A biconvex thin lens is prepared from glass of refractive index $3/2$. The two bounding surfaces have equal radii of 35 cm each. One of the surfaces is silvered to make it reflecting. An object is placed in front of this lens in such a way that the image coincides with the object. If distance (in cm) of object from lens is x , then the value of $2x$ is
8. Two charged particles (m, q) and $(2m, -2q)$ are placed in a gravity free space where a uniform electric field E exists as shown in figure. After the particles are released, they stay at a constant distance from each other. What is the distance between them in centimeter? (Neglect the gravitational interaction).
(K is electrostatic constant and take $\frac{Kq}{E} = \frac{27}{8}$)



9. A circular plane mirror is kept on a horizontal surface, an opaque sphere of radius $R = 20$ cm is placed symmetrically on the circular plane mirror. A small insect lies on the sphere at a vertical height of 36 cm from the circular mirror. Find the minimum radius (in cm) of mirror so that image of insect is formed by the mirror.

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

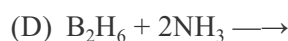
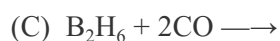
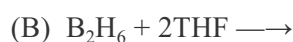
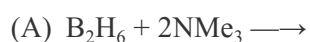
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

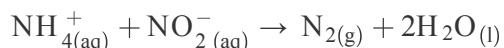
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. In which of the following reaction bond cleavage of B_2H_6 is different from the other reactions.

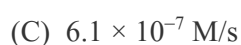
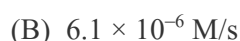
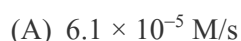


2. Initial rate data at $25^\circ C$ are listed in the table for the reaction



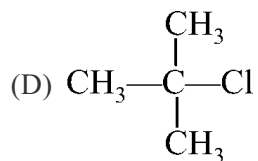
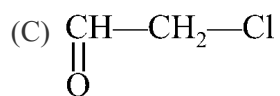
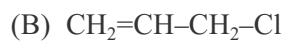
Expt. No.	Initial $[NH_4^+]$	Initial $[NO_2^-]$	Initial rate of consumption of NH_4^+ M/s
1.	0.24	0.10	7.2×10^{-6}
2.	0.12	0.10	3.6×10^{-6}
3.	0.12	0.15	5.4×10^{-6}

What is the reaction rate when concentrations are $[NH_4^+] = 0.39 \text{ M}$ and $[NO_2^-] = 0.052 \text{ M}$?

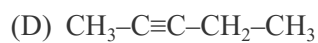
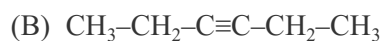
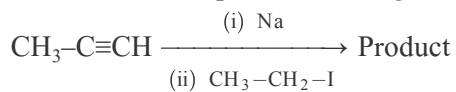


Space for Rough Work

3. Which of the following is most reactive towards S_N2 mechanism



4. What is the final product of the given reaction



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Which of the following statement(s) is/are **CORRECT** -

- (A) $[\text{B}_3\text{O}_6]^{-3}$ ion is non-planar but $\text{B}_3\text{N}_3\text{H}_6$ is planar
- (B) $(\text{H}_3\text{Si})_3\text{N}$ is planar but $(\text{SiH}_3)_3\text{P}$ is pyramidal
- (C) $\text{H}_3\text{C} - \text{NCS}$ is bent but $\text{SiH}_3 - \text{NCS}$ is linear
- (D) $(\text{CH}_3)_3\text{N}$ is pyramidal but $(\text{GeH}_3)_3\text{N}$ is planar

6. Co-ordination entity Wave length of light absorbed (nm)

$[\text{CoCl}(\text{NH}_3)_5]^{2+}$ P

$[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$ Q

$[\text{Co}(\text{NH}_3)_6]^{3+}$ R

$[\text{Co}(\text{CN})_6]^{3-}$ S

Select the correct orders.

(A) $P > R > Q$

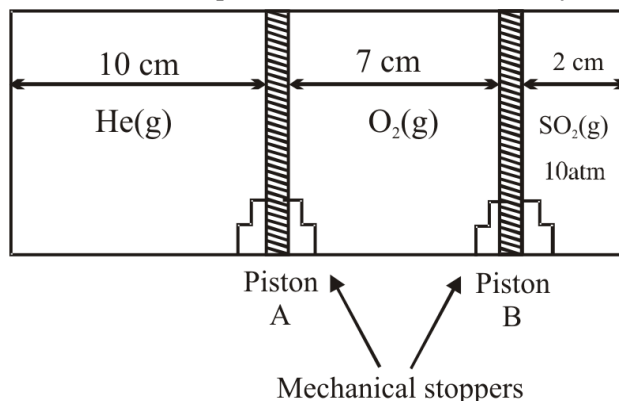
(B) $P > Q > R > S$

(C) $P > Q > R$

(D) $Q > P > R > S$

Space for Rough Work

7. Equal mass of three gases He, O₂ & SO₂ are present in a container initially as shown -



If mechanical stoppers are removed and movable pistons are allowed to attain equilibrium -

Choose the **incorrect** statement -

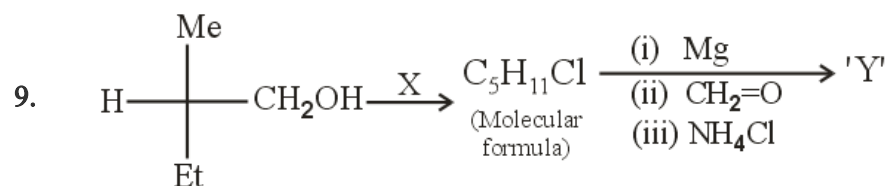
- (A) Position of piston A from the left end of the container is 15 cm
- (B) Position of piston B from the left end of the container is 7cm
- (C) Final pressure of O₂ gas is 10 atm
- (D) Final pressure of He gas is 20 atm

8. For the following reaction

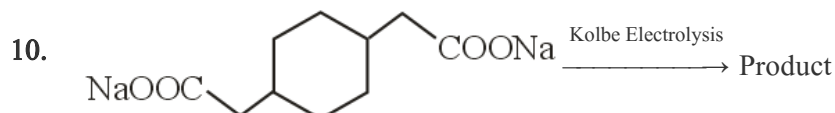
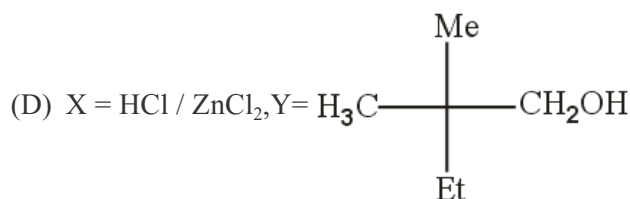
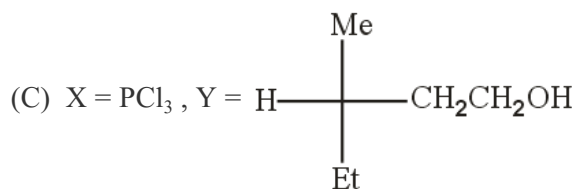
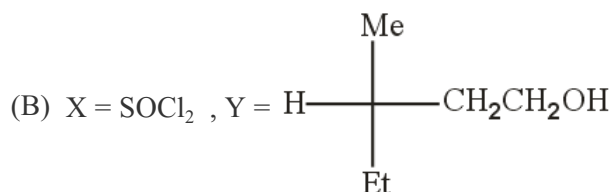
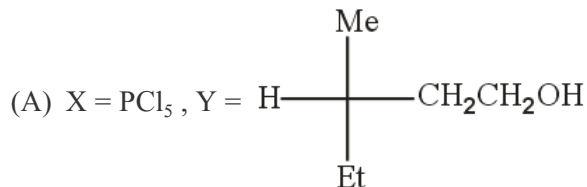
$2\mathbf{X} + \mathbf{Y} \xrightarrow{k} \mathbf{P}$, the rate of reaction is $\frac{d[\mathbf{P}]}{dt} = k[\mathbf{X}]$. Two moles of $\mathbf{X}$ are mixed with one mole of $\mathbf{Y}$ to make 1.0 L of solution. At 50 s, 0.5 mole of $\mathbf{Y}$ is left in the reaction mixture. The correct statement(s) about the reaction is(are) (Use: $\ln 2 = 0.693$)

- (A) The rate constant, k , of the reaction is $13.86 \times 10^{-4} \text{ s}^{-1}$.
- (B) Half-life of $\mathbf{X}$ is 50s.
- (C) At 50 s, $-\frac{d[\mathbf{X}]}{dt} = 13.86 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$.
- (D) At 100 s, $-\frac{d[\mathbf{Y}]}{dt} = 3.465 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$.

Space for Rough Work



Correct pair of X & Y respectively is :



(A) Product contain C_3 axis of symmetry

(B) Product is a bicyclo compound

(C) Product contain plane of symmetry

(D) Product can show stereoisomerism

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

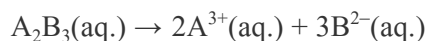
Where two or more than two complexes having similar molecular weight (or) molecular formula but their structural arrangements are different these forms are known as isomer & this phenomenon is known as isomerism; isomerism are of two types structural & stereoisomerism ; stereoisomerism are of two types geometrical isomerism & optical isomerism; compound which has plane of symmetry will not exhibit optical isomerism.

1. Total possible isomers of $K_4[Fe(CN)_6]$.
2. Total geometrical isomers theoretically possible in an octahedral complex of formula $[Co(C_2O_4)(dmg)(en)]$
(In this complex dmg^{-1} is bonded by both 'N' atoms with central metal)

Space for Rough Work

Paragraph for Questions 3 and 4

For the first order dissociation reaction



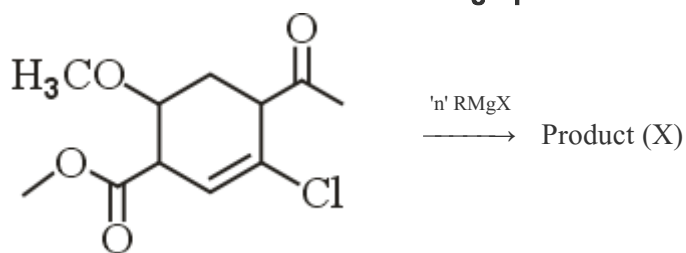
a definite volume of solution is taken at different instant and its osmotic pressure rise is measured.

Osmotic pressure rise at $t = 0$ is 2 mm and at $t = 10$ min is 6 mm.

[Given density of solution is constant and equal to 1 gm/ml, $g = 10 \text{ m/s}^2$]

3. Calculate osmotic pressure of solution in mm at $t = 20$ min.
4. Calculate time (in minute) for 87.5 % of the reaction to complete -

Paragraph for Questions 5 and 6



1 equivalent

5. Value of 'n' is?
6. Degree of unsaturation (D.O.U.) in product (X)?

Space for Rough Work

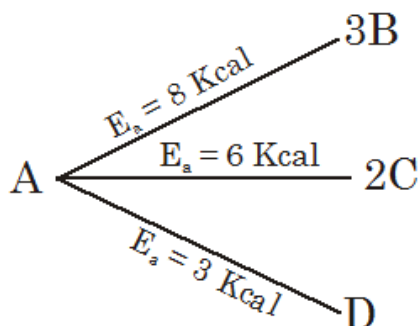
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Consider MOT for B_2 & F_2 molecules & find total number of electrons in both which is/are present in M.O. which is/are symmetrical around bond axis :-
8. A substance undergoes first order parallel reaction as shown. Calculate net activation energy (in kcal) if it is known that B, C and D are formed in a molar ratio of 2 : 3 : 4 respectively.



Express answer in nearest integer

9. For a reaction $A \rightarrow P$
Half life measured for two different values of initial concentrations $5 \times 10^{-3} \text{ M}$ and $25 \times 10^{-4} \text{ M}$ are 1 and 8 hrs respectively. If initial concentration is adjusted to $1.25 \times 10^{-3} \text{ M}$, what is the half-life? (in hrs)

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Value of $\lim_{x \rightarrow 1} \frac{(\log(1+x) - \log 2)(3 \cdot 4^{x-1} - 3x)}{((7+x)^{1/3} - (1+3x)^{1/2}) \sin \pi x}$ is equal to
- (A) $\frac{9}{\pi}(2 \log 2 - 1)$
- (B) $\frac{9}{4\pi}[\log 4 - 1]$
- (C) $\frac{9}{2\pi} \left[\log 4 - \frac{1}{2} \right]$
- (D) $\frac{2}{3\pi}[2 \log 2 - 1]$
2. If $\theta_1, \theta_2, \theta_3$ are three distinct real values such that $\frac{\sin \theta_1 + \sin \theta_2 + \sin \theta_3}{\sin(\theta_1 + \theta_2 + \theta_3)} = \frac{\cos \theta_1 + \cos \theta_2 + \cos \theta_3}{\cos(\theta_1 + \theta_2 + \theta_3)} = 2$ and $\cos(\theta_1 + \theta_2) + \cos(\theta_2 + \theta_3) + \cos(\theta_3 + \theta_1) = \lambda$, then value of $\lim_{x \rightarrow \lambda} \frac{\sqrt{x^2 - \lambda^2}}{\sqrt{x - \lambda} + \sqrt{x} - \sqrt{\lambda}}$ is
- (A) 1 (B) 2 (C) 3 (D) 5

Space for Rough Work

3. If $2f(\sin x) + \sqrt{2}f(\cos x) = \tan x$, then $\lim_{x \rightarrow 1^-} (\sqrt{1-x})f(x)$ is - (Where $x \in \mathbb{R} - \{(2n+1)\frac{\pi}{2}; n \in \mathbb{I}\}$)
- (A) $\frac{1}{2\sqrt{2}}$ (B) $\sqrt{2}$
- (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{2}}{3}$
4. The complete interval of values of x in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying the inequations $\cos x \cdot \cos 2x > -\frac{1}{4}$ and $3\cos x > 1$, is
- (A) $\left(0, \frac{\pi}{3}\right)$
- (B) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$
- (C) $\left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$
- (D) $\left(-\frac{\pi}{3}, \frac{5\pi}{12}\right)$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. If $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{\tan x}{x - \tan x}}$ exists and equal to ℓ , then which of the following is (are) correct ?

(A) $\lim_{x \rightarrow \infty} \left(\frac{2x^2 + 1}{4x^2 - 1} \right)^{\frac{x^2}{2+x}}$ is less than ℓ

(B) If $f_1(x) = \frac{x}{3} + 2 \quad \forall x \in \mathbb{R}$ and $f_n(x)$ is defined by $f_n(x) = f_1(f_{n-1}(x)) \quad \forall n \geq 2$ and $n \in \mathbb{N}$ then $\lim_{n \rightarrow \infty} f_n(x)$ is more than ℓ

(C) $\lim_{x \rightarrow \infty} \left(\frac{2x^2 + 1}{4x^2 - 1} \right)^{\frac{x^2}{2+x}}$ is more than ℓ

(D) If $f_1(x) = \frac{x}{3} + 2 \quad \forall x \in \mathbb{R}$ and $f_n(x)$ is defined by $f_n(x) = f_1(f_{n-1}(x)) \quad \forall n \geq 2$ and $n \in \mathbb{N}$ then $\lim_{n \rightarrow \infty} f_n(x)$ is less than ℓ

Space for Rough Work

6. Which of the following satisfy the equation $\sin^{-1} \sqrt{\frac{1+x}{2}} - \sqrt{2-x} = \cot^{-1} (\tan \sqrt{2-x}) - \sin^{-1} \sqrt{\frac{1-x}{2}}$
- (A) $\left(2 - \frac{\pi^2}{4}, 1\right]$ (B) $\left(2 - \frac{\pi^2}{4}, 0\right)$
- (C) $[-1, 1]$ (D) $\left[1 - \frac{\pi^2}{4}, 1\right]$
7. $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$, $h : \mathbb{R} \rightarrow \mathbb{R}$, defined such that $f \circ g \circ h : \mathbb{R} \rightarrow \mathbb{R}$ is a bijective function then :
- (A) h must be injective
- (B) g may be many one into
- (C) g must be surjective
- (D) f must be surjective
8. A right angled triangle has one leg of length 1, another of length 'x' & hypotenuse of length 'y'. The angle opposite to the side x is θ then -
- (A) $y^n - x^n = \frac{1 - \sin^n \theta}{\cos^n \theta}$
- (B) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y - x) = 0$
- (C) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^2 - x^2) = 1$
- (D) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^3 - x^3)$ does not exist.

Space for Rough Work

9. Which of the following are incorrect ?

- (A) Let f be non constant periodic function with periods (need not be fundamental period) T_1 as well as T_2 then $\frac{T_1}{T_2}$ must be rational
- (B) Let f and g be periodic with periods T_1 and T_2 respectively such that LCM (T_1, T_2) does not exist then $f + g$ must be non periodic
- (C) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an invertible function then $f(x) = x \Leftrightarrow f(x) = f^{-1}(x)$
- (D) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x + y) = f(x) + f(y) \forall x, y \in \mathbb{R}$ then $f(x)$ must be kx for some real number $k \in \mathbb{R}$.

10. The sum of infinite terms of series

$$\cot^{-1} \left(1^2 + \sin^2 \frac{2\pi}{3} \right) + \cot^{-1} \left(2^2 + \sin^2 \frac{4\pi}{3} \right) + \cot^{-1} \left(3^2 + \sin^2 \frac{8\pi}{3} \right) + \dots \text{ is}$$

- (A) $\cot^{-1}(1)$
- (B) $\tan^{-1}(2)$
- (C) $\frac{1}{2} \sin^{-1} \left(\frac{4}{5} \right)$
- (D) $\cot^{-1}(1) + \cot^{-1}(3)$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \min \{f(t) ; 0 \leq t \leq x\} & ; 0 \leq x \leq 1 \\ 3 - x & ; 1 < x \leq 2 \end{cases}$, then

1. $g\left(\frac{1}{4}\right)$ is equal to
2. $g\left(\frac{3}{4}\right)$ is equal to

Space for Rough Work

Paragraph for Questions 3 and 4

Let α be a real root of $f(x) = 8x^3 + 12x^2 + 6x + 1$, then

3. The value of $\sin^{-1}(\sin[\alpha]) + \cos^{-1}(\cos[\alpha])$ is equal to
4. The number of distinct real roots of equation $f(x) = 0$, is
($[.]$ denotes the greatest integer function)

Paragraph for Questions 5 and 6

Let $a_1 = 1$ and $a_n = \sin(a_{n-1})$, $n > 1$ & $n \in \mathbb{N}$.

If $\lim_{n \rightarrow \infty} \frac{2^{2a_n} - 2^{1+a_n} \cdot 3^{a_n} + 3^{2a_n}}{\cos a_n + 1 - e^{a_n} - e^{-a_n}} = -\frac{\alpha}{3} \ln \beta \left(\frac{2}{3} \right)$, (where α and β are prime numbers).

5. The value of 3α is
6. The value of 2β is

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. If a sequence is such that $a_1 = 1$ and $a_n = n(1 + a_{n-1}) \forall n \geq 2$ and $n \in \mathbb{N}$,
then $\lim_{n \rightarrow \infty} \ln \left(\left(1 + \frac{1}{a_1}\right) \left(1 + \frac{1}{a_2}\right) \dots \left(1 + \frac{1}{a_n}\right) \right)$ is _____.
8. If the solution set of $\sin^{-1}x + \sin^{-1}y \geq (x+y)^2 - 2[x(y+1) - (1-y)] + \pi$ is (x_0, y_0) then $3x_0 + y_0$ is
9. If $\lim_{x \rightarrow 0} \frac{a + \log_e(1+x) + b \sin x}{x^2} = \ell$ (finite) then $|a + b + 4\ell|$ (where $a, b, \ell \in \mathbb{R}$)

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work